

for relu_max , $\text{masc}(z) = \max(0, z)$

for P_2 , if $t > 0$, it equals $\frac{b}{b + \text{relu}(-t + b)}$

if $t < b$, $P_2 = \frac{b}{2b - t}$

if $t \geq b$, $P_2 = 1$

$t \leq 0$: $P_2 = \frac{\text{relu}(t + b)}{b + \text{relu}(t + b)}$

if $t \leq -b$, $P_2 = 0$

if $t > -b$, $P_2 = \frac{t + b}{2b + t}$

derivative: $t \geq b$ or $t \leq -b$: 0

$0 < t < b$: $\frac{b}{(2b - t)^2}$

$-b < t \leq 0$: $\frac{1}{(2b + t)^2}$

$\text{sparse_max}(z)_i = \text{relu}(z_i + c - \tau_i')$
 since τ has to be such ~~that~~ as to satisfy normaliza-
 tion of writing $\text{relu}(z_i' - \tau_i')$ as $\text{max}(0, z_i' - \tau_i')$:

we have to have $\{ \text{max}(0, z_i + c - \tau_i') = 1 \}$

notice $\tau_i' = \tau_i + c$ gives us the normalization
 constraint for z , and since τ is ~~unique~~

$$\begin{aligned}
 \tau_i' &= \tau_i + c \\
 \text{so } \text{max}(0, z_i' - \tau_i') &= \text{max}(0, z_i + c - (\tau_i + c)) \\
 &= \text{sparse_max}(z)_i
 \end{aligned}$$

the unique scalar
 that satisfies this condition

$$\lim_{T \rightarrow 0^+} \text{soft_max}\left(\frac{z}{T}\right)_i = \lim_{T \rightarrow 0^+} \frac{e^{\frac{z_i}{T}}}{\sum_j e^{\frac{z_j}{T}}}$$

in the limit, when $z_j - z_i < 0$
 the i th component is 0, so
 for the arg max z_i , all
 components except the ~~one~~ i th are 0, the ~~other~~ k th
 being 1 ~~cause~~ due to $e^0 = 1$ k th

$$3.4 - \text{relevance}(z) = \frac{\sum_i m_i}{S}$$

$$S = \sum m_i$$

We will say that $m_i \in S$ if $z_i - z_k + b > 0$,
and $m_i > 0$

We will have a quotient rule:

$$J_{ij} = \frac{\partial \text{relevance}(z)}{\partial z_j} = \frac{S \cdot \frac{\partial m_i}{\partial z_j} - m_i \frac{\partial S}{\partial z_j}}{S^2}$$

$$= \frac{1}{S} \left(\frac{\partial m_i}{\partial z_j} - \text{relevance}(z) \frac{\partial S}{\partial z_j} \right)$$

if $i \notin S$ $m_i = 0$; m_i is also zero and

$$\frac{\partial m_i}{\partial z_j} = 0 \quad J_{ij} = 0 \quad i \notin S$$

now $i \in S$, and $i \neq k$:

$$\frac{\partial m_i}{\partial z_j} = G_{ij} - G_{kj}$$

$$\text{and } \frac{\partial S}{\partial z_j} = 1$$

$$\text{sparsenorm}(z) = \arg \min_{p \in \Delta_K} \|p_i - z_i - c\|^2$$

say $p_i = z_i + a$. Expanding the sum of squares and substituting a , we obtain

$$\arg \min_{p \in \Delta_K} (\|p_i - z_i\|^2 - 2c \sum_i (p_i - z_i) + \sum_i c^2)$$

since $\sum_i c$

$$\text{relu}_{\max, b}(z) = \frac{\text{relu}(z_i + c - \max(z) + b)}{\sum_i (z_i + c - \max(z) + b)} \quad (7)$$

$\max(z + c) = \max(z) + c$, since c is ~~not~~ constant

so (7) turns into

$$\text{relu}(z_i + c - \max(z) + b)$$

$$\sum_i (z_i + c - \max(z) + b)$$

$$= \text{relu}_{\max, b}(z)$$

for sparseness, for P_i to be non zero, ~~not~~,
from ~~the~~ the relu function we ~~get~~ know that

$$\frac{z_{k,i}}{\tau} > \tau \quad (1)$$

now we also know that some ~~of~~ z_i must
be greater than zero, and a probability compo-
nent is at most 1, so $\frac{z_i}{\tau} - \tau \leq 1 \quad (2)$

now, with the intention to move towards the
desired solution, which could be ~~also~~ obtained
otherwise ~~in~~ numerically/computationally,
we take z_i in (1) to be $\neq z_k$, and z_i in
(2) to be z_k , the maximum component of the
vector z , and combining both equations:

$$\frac{z_k - 1}{\tau} \leq \tau \leq \frac{z_i}{\tau} \quad \text{so} \quad \frac{z_k - 1}{\tau} \leq z_i$$

$$\Rightarrow \underbrace{z_k - z_i}_{>0} \leq \underbrace{\tau}_{\rightarrow 0^+}$$

so in the limit the condition doesn't hold,
so (1) doesn't hold, by contradiction.

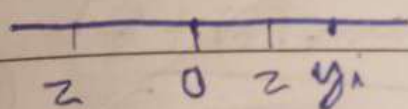
Notice only the limit being 0^+ guarantees any
 z_i is zero, ~~not~~ and also that

3.1-1-

Notice $\|y - z\| = \left(\sum_{i=1}^k |y_i - z_i| \right)^{\frac{1}{2}}$

~~Since $(y_i - z_i)^2$~~ to minimize $\|y - z\|^2$ we need to obtain the minimum of each part of the sum, $\|y_i - z_i\| = |y_i - z_i|$

~~with~~ with the only constraint that $y_i \geq 0$, when $z \leq 0$ argmin $\|y - z\|$ is $y = 0$



whereas if $z \geq 0$, y can be equal to

z , which minimizes the distance between them.

Thus we obtain the result, assuming $z \geq 0$ means $z_i \geq 0 \quad i \in \{1, \dots, k\}$

3.2- for softmax:

$$\text{softmax}_i = \frac{e^{z_i + c}}{\sum_{j=1}^k e^{z_j + c}} = \frac{e^c}{e^c} \left(\frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}} \right) = \text{softmax}(z)_i$$

$$\text{if } j=i: \frac{\partial m_i}{\partial z_i} = 1 \quad \frac{\partial S}{\partial z_i} = 1$$

$$J_{ii} = \frac{1}{S} (1 - \text{relevance}(z)_i)$$

$$\text{if } j \neq i \text{ and } j \neq k: \frac{\partial m_i}{\partial z_j} = 0 \quad \frac{\partial S}{\partial z_j} = 1 \quad J_{ij} = \frac{-\text{relevance}(z)_i}{S}$$

$$\text{now } i \neq j \text{ but } j=k: \frac{\partial m_i}{\partial z_k} = -1 \quad \frac{\partial S}{\partial z_k} = 1 - |S(z)|$$

$$J_{ik} = \frac{1}{S} (-1 - \text{relevance}(z)_i (1 - |S(z)|))$$

$$\text{now } i=k: \frac{\partial m_k}{\partial z_j} = 0 \text{ because } m_k = k$$

$$\text{if } j \neq k: \frac{\partial S}{\partial z_j} = 1 \quad J_{kj} = \frac{-\text{relevance}(z)_k}{S}$$

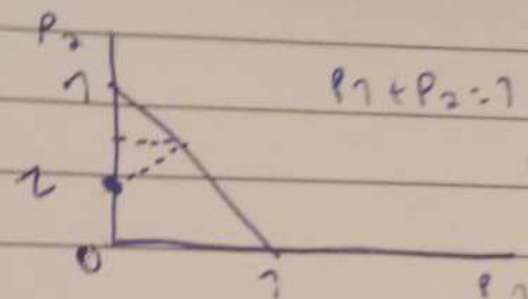
$$\text{if } j=k: \frac{\partial S}{\partial z_k} = 1 - |S(z)| \quad J_{kk} = \frac{\text{relevance}(z)_k}{S} (1 - |S(z)|)$$

3.3.

$$z_{\text{softmax}}(z) = \frac{e^{\frac{z}{0.4+2^t}}}{e^{\frac{z}{0.4+2^t}} + 1} = G(t)$$

$$\frac{\partial}{\partial t} G(t) = G(t)(1 - G(t))$$

$\text{softmax}_2(z)$:



$p_1 + p_2 = 1$ is the constraint line

we just need to project $z = (0, t)$ onto the line orthogonally, to

minimize distance.

if $t \geq 1$, obviously $\text{softmax}_2(z) = (0, 1)$

if $t \leq -1$, $\text{softmax}_2(z) = (1, 0)$

if $|t| < 1$, $\text{softmax}_2(z) = \left(\frac{1-t}{2}, \frac{t+1}{2}\right)$

p_2 is $\frac{t+1}{2}$, visually, it's the average

between t and 1 , as shown in the graph

and $p_1 = 1 - p_2$

the derivative is 0, 5 when $|t| < 1$ and 0 otherwise

every component of z being different is also necessary

$$\text{relumasc}_g(z)_i = \frac{\text{relu}(z_i - z_k + b)}{T}$$

$$\sum_j \text{relu}(z_j - z_k + b)$$

$$\text{for } i \neq k: \frac{z_i - z_k - 0 - \infty}{T}$$

so the denominator goes to zero, and the denominator to b with the term in the sum which gives $\frac{\text{relu}(z_k - z_k + b)}{T}$

$i = k$ makes the numerator b , and we get the desired result

for $\text{relumasc}_g(z)_i$ to be equal to $\text{sparsenasc}_g(z)_i$ we simply take the denominator of the latter (the normalization) to be one, and equal the arguments of the two relu's, obtaining:

$$z_i - \gamma = z_i - z_k + b \quad \text{and} \quad b = z_k + \gamma$$

and the denominator of $\text{relumasc}_g(z)_i$ becomes $\sum_j (z_j - \gamma)$ which is 1 due to sparsenasc's definition

3.5 - Both sparseness and allness / as seen by the plots, produce vanishing gradients, leading to numerical explosion, and are "less" smooth and differentiable than softmax, which can cause the same kinds of issues.

$$L = -\log(P_i + \epsilon) + \log(C)$$

$$\frac{\partial L}{\partial z} = \frac{-1}{P_i + \epsilon} (\nabla_z P_i)$$

where $P_i = \text{softmax}(z)_i$
and thus $\nabla_z P_i$ is the ith row of the Jacobian

$$\frac{\partial L}{\partial z_j} = \frac{-1}{P_i + \epsilon} J_{ij}$$